

Sarah Allec

Materials Scientist, Materials Discovery Team · Pacific Northwest
National Laboratory

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[GitHub](#) · [LinkedIn](#) · [Google Scholar](#)
· [ORCID](#)

I combine physics-based simulation with AI, from Bayesian models to autonomous agents, to design new materials and close the loop between computation and experiment.

Experience

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|----------------|---|
| 2024 – present | Materials Scientist
Pacific Northwest National Laboratory · Materials Discovery Team · Richland, WA
Research on active learning, uncertainty-aware machine learning, and atomistic modeling for materials discovery, autonomous synthesis, and nuclear waste form design. |
| 2022 – 2024 | Research Scientist II
Citrine Informatics · External Research Department · Remote
Materials informatics research on funded programs, including physics-informed machine learning for glass design and multi-institutional data management for combinatorial materials science. |
| 2020 – 2022 | Postdoctoral Research Associate
Pacific Northwest National Laboratory · Basic & Applied Molecular Foundations Group · Richland, WA
Computational design of carbon capture solvents and catalysts using ab initio molecular dynamics and density functional theory. |

Education

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| 2015 – 2020 | Ph.D. in Materials Science & Engineering
University of California, Riverside <ul style="list-style-type: none">• Thesis: Atomistic Modeling of Amorphous Materials• Advisors: P. Alex Greaney (Ph.D.) and Bryan M. Wong (M.S., 2018)• NSF Graduate Research Fellowship, NASA MIRO FIELDS Graduate Student Fellowship |
| 2011 – 2015 | B.S. in Applied Mathematics (Physics emphasis)
University of California, Riverside <ul style="list-style-type: none">• Graduated summa cum laude• Regents Scholarship |

Publications 29 peer-reviewed articles and book chapters

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| 2026 | 29. Predicting Nepheline Formation in Nuclear Waste Glasses Using Machine Learning With Uncertainty Quantification M. Diaz-Acevedo, C. E. Curry, X. Lu, S. Chong, S. I. Allec , J. D. Vienna. <i>International Journal of Applied Glass Science</i> 17 , e70062 (2026). doi:10.1111/ijag.70062 |
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28. **Boron Coordination in Multicomponent Glasses: Analytical Models and Machine Learning With Uncertainty** C. E. Curry, M. Diaz-Acevedo, D. Wang, S. I. Allec, J. J. Neeway, J. D. Vienna, X. Lu. *International Journal of Applied Glass Science* **17**, e70043 (2026). doi:10.1111/ijag.70043
 27. **Toward Intelligent Multimodal Holography for Real-Time Chemical Imaging of Dynamic Ion Separation** G. Ricchiuti, S. I. Allec, M. Ziatdinov, V. Prabhakaran. *Advanced Intelligent Discovery* **2**, e202500237 (2026). doi:10.1002/aidi.202500237
 26. **Comparison of DeePMD, MTP, GAP, ACE and MACE Machine-Learned Potentials for Radiation-Damage Simulations: A User Perspective** A. Roy, R. Devanathan, S. Allec, G. Nandipati, A. M. Casella, D. J. Senior, D. D. Johnson, G. Balasubramanian, A. Souلامي. *Advanced Intelligent Discovery*, e202500196 (2026). doi:10.1002/aidi.202500196
 25. **Agentic workflow enables the recovery of critical materials from complex feedstocks via selective precipitation** A. Ritchhart, S. I. Allec, P. Butreddy, K. Kulesa, Q. Wang, D. T. Nguyen, M. Ziatdinov, E. Nakouzi. *Materials Horizons* **13**, 7008-7014 (2026). doi:10.1039/d6mh00475j
 24. **Assessing universal MLIP robustness with per-atom uncertainty for simulations of solid-liquid interfaces** J. A. Bilbrey, J. S. Firoz, S. I. Allec, H. W. Sprueill, A. D. von Rueden, B. A. Jackson, S. Rauei, M. S. Lee, S. Choudhury. *npj Computational Materials* **12**, 291 (2026). doi:10.1038/s41524-026-02051-8
- 2025**
23. **Toward autonomous materials synthesis via reaction-diffusion coupling** A. Ritchhart, S. I. Allec, H. M. Job, P. Butreddy, M. Ziatdinov, E. Nakouzi. *APL Machine Learning* **3**, 036109 (2025). doi:10.1063/5.0267949
 22. **Active and transfer learning with partially Bayesian neural networks for materials and chemicals** S. I. Allec, M. Ziatdinov. *Digital Discovery* **4**, 1284-1297 (2025). doi:10.1039/d5dd00027k
 21. **Computational Investigation of a CO₂ Conversion Strategy via Diels-Alder Reaction in a Carbon Capture Solvent** D. Zhang, M. T. Manetsch, S. I. Allec, L. Kollias, R. Rousseau, D. J. Heldebrant, V. A. Glezakou. *ACS Omega* **10**, 23663-23672 (2025). doi:10.1021/acsomega.5c02620
- 2024**
20. **A Case Study of Multimodal, Multi-institutional Data Management for the Combinatorial Materials Science Community** S. I. Allec, E. S. Muckley, N. S. Johnson, C. K. H. Borg, D. J. Kirsch, J. Martin, R. Pant, I. Takeuchi, A. S. Lee, J. E. Saal, L. Ward, A. Mehta. *Integrating Materials and Manufacturing Innovation* **13**, 406-419 (2024). doi:10.1007/s40192-024-00345-7
 19. **Molecular Understanding of Nitrogen Oxide Fixation of Water-Lean Carbon Capture Solvents by Atomistic Modeling** L. Kollias, M. T. Nguyen, S. I. Allec, D. Malhotra, D. Zhang, R. Rousseau, V. A. Glezakou, P. K. Koech, D. J. Heldebrant. *Industrial & Engineering Chemistry Research* **63**, 12316-12324 (2024). doi:10.1021/acs.iecr.4c01143

18. **Evaluation of GlassNet for physics-informed machine learning of glass stability and glass-forming ability** S. I. Allec, X. Lu, D. R. Cassar, X. T. Nguyen, V. I. Hegde, T. Mahadevan, M. Peterson, J. Du, B. J. Riley, J. D. Vienna, J. E. Saal. *Journal of the American Ceramic Society* **107**, 7784-7799 (2024). doi:10.1111/jace.19937
17. **Tetrameric self-assembling of water-lean solvents enables carbamate anhydride-based CO₂ capture chemistry** J. Leclaire, D. J. Heldebrant, K. Grubel, J. Septavaux, M. Hennebelle, E. D. Walter, Y. Chen, J. L. Bañuelos, D. Zhang, M. T. Nguyen, D. Ray, S. I. Allec, D. Malhotra, W. Joo, J. King. *Nature Chemistry* **16**, 1160-1168 (2024). doi:10.1038/s41557-024-01495-z
16. **Towards informatics-driven design of nuclear waste forms** V. I. Hegde, M. Peterson, S. I. Allec, X. Lu, T. Mahadevan, T. Nguyen, J. Kalahe, J. Oshiro, R. J. Seffens, E. K. Nickerson, J. Du, B. J. Riley, J. D. Vienna, J. E. Saal. *Digital Discovery* **3**, 1450-1466 (2024). doi:10.1039/d4dd00096j

2023

15. **The Role of Surface Hydroxyls in the Mobility of Carboxylates on Surfaces: Dynamics of Acetate on Anatase TiO₂(101)** R. Ma, C. R. O'Connor, G. Collinge, S. I. Allec, M. S. Lee, Z. Dohnálek. *The Journal of Physical Chemistry Letters* **14**, 2542-2550 (2023). doi:10.1021/acs.jpclett.3c00175
14. **Dynamic Evolution of Palladium Single Atoms on Anatase Titania Support Determines the Reverse Water-Gas Shift Activity** L. Chen, S. I. Allec, M. T. Nguyen, L. Kovarik, A. S. Hoffman, J. Hong, D. Meira, H. Shi, S. R. Bare, V. A. Glezakou, R. Rousseau, J. Szanyi. *Journal of the American Chemical Society* **145**, 10847-10860 (2023). doi:10.1021/jacs.3c02326
13. **Enhancing CO₂ Transport Across a PEEK-Ionene Membrane and Water-Lean Solvent Interface** E. D. Walter, D. Zhang, Y. Chen, K. S. Han, J. D. Bazak, S. Burton, K. O'Harra, D. W. Hoyt, J. E. Bara, D. Malhotra, S. I. Allec, V. A. Glezakou, D. J. Heldebrant, R. Rousseau. *ChemSusChem* **16**, e202300157 (2023). doi:10.1002/cssc.202300157

2022

12. **Advanced Theory and Simulation to Guide the Development of CO₂ Capture Solvents** L. Kollias, D. Zhang, S. I. Allec, M. T. Nguyen, M. S. Lee, D. C. Cantu, R. Rousseau, V. A. Glezakou. *ACS Omega* **7**, 12453-12466 (2022). doi:10.1021/acsomega.1c07398
11. **Assessing entropy for catalytic processes at complex reactive interfaces** L. Kollias, G. Collinge, D. Zhang, S. I. Allec, P. K. Gurunathan, G. Piccini, S. F. Yuk, M. T. Nguyen, M. S. Lee, V. A. Glezakou, R. Rousseau. *Annual Reports in Computational Chemistry*, 3-51 (2022). Book chapter. doi:10.1016/bs.arcc.2022.09.004

2021

10. **Amphiphilic Water-Lean Carbon Capture Solvent Wetting Behavior through Decomposition by Stainless-Steel Interfaces** M. T. Nguyen, K. Grubel, D. Zhang, P. K. Koech, D. Malhotra, S. Allec, R. Rousseau, V. A. Glezakou, D. J. Heldebrant. *ChemSusChem* **14**, 5283-5292 (2021). doi:10.1002/cssc.202101350

9. **The electrolyte comprising more robust water and superhalides transforms Zn-metal anode reversibly and dendrite-free** C. Zhang, W. Shin, L. Zhu, C. Chen, J. C. Neufeind, Y. Xu, **S. I. Allec**, C. Liu, Z. Wei, A. Daniyar, J. Jiang, C. Fang, P. A. Greaney, X. Ji. *Carbon Energy* **3**, 339-348 (2021). doi:10.1002/cey2.70
8. **The role of sub-surface hydrogen on CO₂ reduction and dynamics on Ni(110): An ab initio molecular dynamics study** S. I. Allec, M. T. Nguyen, R. Rousseau, V. A. Glezakou. *The Journal of Chemical Physics* **155**, 044702 (2021). doi:10.1063/5.0048894

2019

7. **Heterogeneous CPU+GPU-Enabled Simulations for DFTB Molecular Dynamics of Large Chemical and Biological Systems** S. I. Allec, Y. Sun, J. Sun, C. A. Chang, B. M. Wong. *Journal of Chemical Theory and Computation* **15**, 2807-2815 (2019). doi:10.1021/acs.jctc.8b01239
6. **Linear-Response and Real-Time, Time-Dependent Density Functional Theory for Predicting Optoelectronic Properties of Dye-Sensitized Solar Cells** S. I. Allec, A. Kumar, B. M. Wong. *Dye-Sensitized Solar Cells*, 171-201 (2019). Book chapter. doi:10.1016/B978-0-12-814541-8.00005-7
5. **Chirality Induced Spin Selectivity of Photoexcited Electrons in Carbon-Sulfur [n]Helicenes** J. M. Matxain, J. M. Ugalde, V. Mujica, **S. I. Allec**, B. M. Wong, D. Casanova. *ChemPhotoChem* **3**, 770-777 (2019). doi:10.1002/cptc.201900128

2018

4. **A Highly Stretchy, Transparent Elastomer with the Capability to Automatically Self-Heal Underwater** Y. Cao, H. Wu, **S. I. Allec**, B. M. Wong, D. S. Nguyen, C. Wang. *Advanced Materials* **30**, 1804602 (2018). doi:10.1002/adma.201804602

2017

3. **A Transparent, Self-Healing, Highly Stretchable Ionic Conductor** Y. Cao, T. G. Morrissey, E. Acome, **S. I. Allec**, B. M. Wong, C. Keplinger, C. Wang. *Advanced Materials* **29**, 1605099 (2017). doi:10.1002/adma.201605099

2016

2. **Inconsistencies in the Electronic Properties of Phosphorene Nanotubes: New Insights from Large-Scale DFT Calculations** S. I. Allec, B. M. Wong. *The Journal of Physical Chemistry Letters* **7**, 4340-4345 (2016). doi:10.1021/acs.jpclett.6b02271
1. **Unusual Bandgap Oscillations in Template-Directed π -Conjugated Porphyrin Nanotubes** S. I. Allec, N. V. Ilawe, B. M. Wong. *The Journal of Physical Chemistry Letters* **7**, 2362-2367 (2016). doi:10.1021/acs.jpclett.6b01020

Open-source software

SciLink (core contributor) · github.com/ziatdinovmax/SciLink

An open-source platform of LLM-powered agents that plan experiments, analyze multimodal data, and run simulations. I build the simulation agents, which set up and run DFT and molecular dynamics calculations and recommend structures.

SciLink Tutorials (author) · github.com/sarah-allec/SciLink-Tutorials

Hands-on notebooks for SciLink from the MRS Spring 2026 tutorial and the MEM-C Seed AI Fellowship, covering knowledge-grounded planning, Bayesian optimization, and active-learning DFT screening.

RDsynthesisML (author) · github.com/sarah-allec/RDsynthesisML

Scripts and notebooks reproducing "Toward autonomous materials synthesis via reaction-diffusion coupling", from image scalarization through Bayesian optimization of the next experiment.

Invited talks

Advancing Autonomous Materials Research with Uncertainty-Aware AI and Human-AI Collaboration. Materials Research Society (MRS) Spring Meeting, Honolulu, HI, 2026.

Novel approaches to informatics-driven nuclear waste form design: dataset curation, surrogate modeling, and sequential learning. ACerS Glass & Optical Materials Division (GOMD) Meeting, 2024.

Tutorials

Deploying Agentic AI in Materials Characterization Workflows (MT01). Presented "The Agent Toolkit: Hybrid Models, Active Learning, and Theory-in-the-Loop". Materials Research Society (MRS) Spring Meeting, Honolulu, HI, 2026.

Awards & fellowships

NERSC Director's Reserve Award

National Energy Research Scientific Computing Center, 2024. Active machine learning for nuclear waste form design

DOE Mission Science Award

National Energy Research Scientific Computing Center, 2022 – 2023. Computational screening of redox couples for electrochemical direct air capture of CO₂

NSF Graduate Research Fellowship

National Science Foundation, 2017 – 2020

NASA MIRO FIELDs Graduate Student Fellowship

NASA, 2016 – 2017

Regents Scholarship

University of California, Riverside, 2011 – 2015

Mentoring & service

Peer reviewer

RSC Digital Discovery; DOE Office of Science SBIR/STTR proposals, 2023 – present

Undergraduate research mentor

Cal Poly San Luis Obispo Materials Engineering Summer Research Program, 2023

Workshop leader

North American Materials Education Symposium, 2023

Community volunteer

Interfaith Sanctuary, Boise, ID, 2023 – present

Graduate intern mentor

DOE Science Undergraduate Laboratory Internships (SULI) program, PNNL, 2021

Co-President and Treasurer

Association for Women in Science, UC Riverside chapter, 2018 – 2020

Mentor and tutor

School on Wheels, Riverside, CA, 2018 – 2021